

Algebraic Representation of the Zebra Program

Algebraic Operators

We assume the following algebraic operators:

- $\circ$: sequential composition of constraints
- $\otimes$: parallel (independent) composition
- $\oplus$: nondeterministic choice
- μ : least fixpoint (recursion / search)

Primitive constraint generators:

$\text{unify}(\cdot), \text{mem}(\cdot), \text{shift}, \text{swap}$

Definition of `iright`

The right-adjacency relation is defined as a recursive constraint generator:

$$\text{ir} := \mu X. \left(\text{match}(L, R) \oplus \text{shift} \circ X \right)$$

Definition of `nextto`

The symmetric adjacency relation is defined by disjunction:

$$\text{nt} := \text{ir} \oplus \text{swap} \circ \text{ir}$$

Algebraic Form of the Zebra Program

The Zebra puzzle is represented as a single fixpoint expression:

$$\begin{aligned}
 Z := & \mu \left(\text{unify} \left([\text{house}(\text{norwegian}, -, -, -, -), -, \text{house}(-, -, -, \text{milk}, -), -, -] \right) \right. \\
 & \circ \bigotimes \left(\text{mem}(\text{house}(\text{englishman}, -, -, -, \text{red})) \right. \\
 & \quad \otimes \text{mem}(\text{house}(\text{spaniard}, \text{dog}, -, -, -)) \\
 & \quad \otimes \text{mem}(\text{house}(-, -, -, \text{coffee}, \text{green})) \\
 & \quad \otimes \text{mem}(\text{house}(\text{ukrainian}, -, -, \text{tea}, -)) \\
 & \quad \otimes \text{mem}(\text{house}(-, \text{snails}, \text{winston}, -, -)) \\
 & \quad \otimes \text{mem}(\text{house}(-, -, \text{kools}, -, \text{yellow})) \\
 & \quad \otimes \text{mem}(\text{house}(-, -, \text{luckystrike}, \text{oj}, -)) \\
 & \quad \otimes \text{mem}(\text{house}(\text{japanese}, -, \text{parliaments}, -, -)) \\
 & \quad \otimes \text{mem}(\text{house}(W, -, -, \text{water}, -)) \\
 & \quad \otimes \text{mem}(\text{house}(Z, \text{zebra}, -, -, -)) \\
 & \circ \left(\text{ir}(\text{house}(-, -, -, -, \text{ivory}), \text{house}(-, -, -, -, \text{green})) \right. \\
 & \quad \otimes \text{nt}(\text{house}(-, -, \text{chesterfield}, -, -), \text{house}(-, \text{fox}, -, -, -)) \\
 & \quad \otimes \text{nt}(\text{house}(-, -, \text{kools}, -, -), \text{house}(-, \text{horse}, -, -, -)) \\
 & \quad \left. \left. \left. \otimes \text{nt}(\text{house}(\text{norwegian}, -, -, -, -), \text{house}(-, -, -, -, \text{blue})) \right) \right) \right)
 \end{aligned}$$

Interpretation

This single expression represents the complete Zebra program. Execution corresponds to projection into a lower-dimensional operational semantics (e.g. Prolog), while refactoring and optimization correspond to algebraic transformation at this level.